



Master's Degree in Cognitive Science

NEURAL MASS AND SPIKING MODELS FOR THE STUDY OF NEURAL CIRCUITS

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Abstract

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Chapter 1

Stochastic Resonance in the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1–3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

1.1.1 Classical stochastic resonance

The idea of studying the effects of noise on the dynamics of a system first appeared parallely in the work of Benzi et al. [5] and Nicolis [6], where the term Stochastic Resonance (SR) was used to explain the periodic behavior of earth's average temperature.

A brief covering of that first SR application will be made, as it provides a practical understanding of the general phenomenon and a surprising link to neural systems.

The most common way to model temperature averages was using simple energy balance deterministic models:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = R_{in}(T) - R_{out}(T) \tag{1.1}$$

where C is the earth's thermal capacity, and R_{in} and R_{out} represent the total incoming and outgoing solar radiation, defined as

$$\begin{aligned} R_{in} &= Q\mu \\ R_{out} &= \alpha(T)Q\mu + \varepsilon(T) \end{aligned}$$

where μ is the average incoming solar radiation.

In [5] a simplification is used: 3 steady-states are assumed to exist as $T_1 < T_2 < T_3$, those are possible earth's climates. 1.1 then becomes

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mu\varepsilon(T)}{1 + \beta\,(1 - T/T_1)\,(1 - T/T_2)\,(1 - T/T_3)} - \varepsilon(T) \tag{1.2}$$

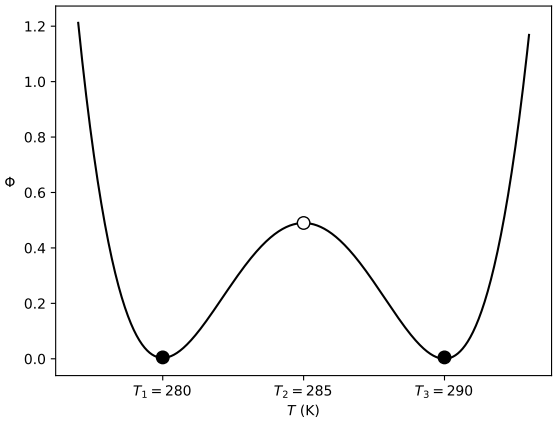


Figure 1.1: The double well potential. Filled circles are stable points.

To analyze the stability of the defined climates the rate of change of temperatures is taken as

$$F(T) = \frac{\mathrm{d}T}{\mathrm{d}t} \ , \tag{1.3}$$

then a pseudo-potential energy function is defined as

$$\Phi(T) = -\int F(T)\mathrm{d}T \tag{1.4}$$

that represents the "energy" needed to change the temperature across temperature values. From Φ we get

$$F(T) = -\frac{\partial \Phi}{\partial T} \tag{1.5}$$

Then intuitively the maximum of Φ will be an unstable point, while the minima will be stable. That is, where the second derivative with respect to temperature

$$\frac{\mathrm{d}F}{\mathrm{d}T} = -\frac{\partial^2 \Phi}{\partial T^2} \tag{1.6}$$

is < 0, any neighboring starting state of the earth's temperature will converge to it, while when it is > 0 they will diverge.

It can be shown that T_1, T_3 are stable points, while T_2 is unstable: a peak any starting state would "roll off" from (see A). In our context, T_1, T_3 are the only observable climates, temperatures at which the earth will stay for enough time to be detectable in the geological records. A system where 2 stable states separated by an unstable state exist is called a bistable system, and the plot of Φ over T displays the typical double well shape of the (pseudo)potential function (Figure 1.1).

With the present model, we can see that the earth's climate would depend solely (all else being equal) on the starting temperature, i.e., whether it is smaller or greater than T_2. Then it would follow $F(T)$ towards T_1 or T_3 (Figure 1.2). However, earth's temperatures do indeed show oscillations with a period of 10^5 years.

A weak cosinusoidal forcing modulating the input solar radiations can be introduced, representing the varying distance of the earth from the sun along its elliptic orbit. Then μ becomes time dependent

$$\mu(t) = 1 + \gamma \cos(\omega t) \tag{1.7}$$

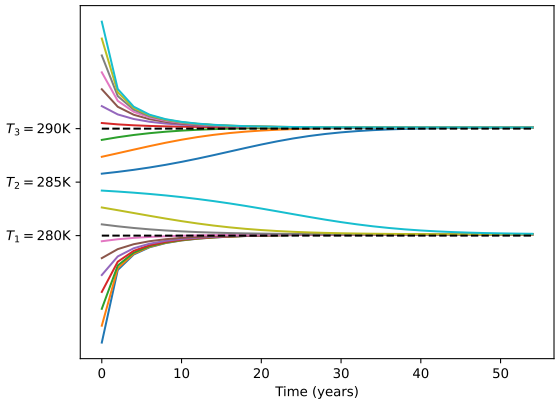


Figure 1.2: Trajectories for different starting temperatures without noise.

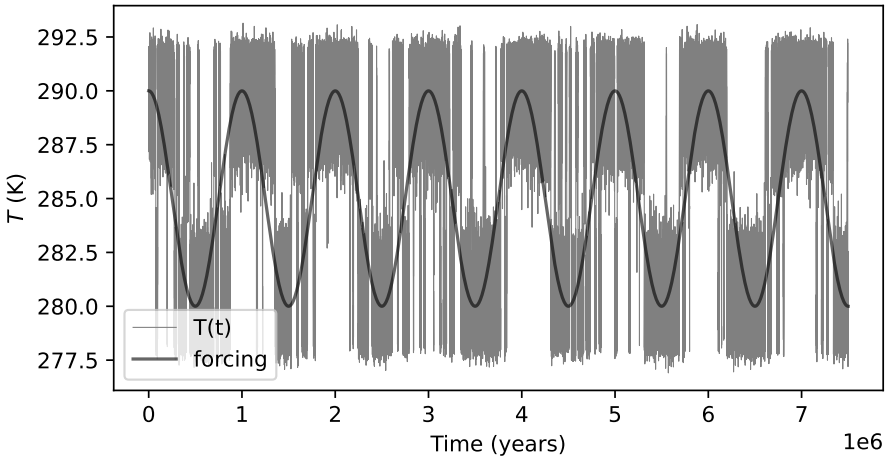


Figure 1.3: Temperature over time from numerical solution of 1.2, with $\sigma = 0.15$.

where $\gamma \ll 1$ and $\omega = 2\pi/10^5$, reflecting the observed oscillation period. The derivative and pseudo-energy function become time dependent $\tilde{F} = \tilde{F}(T;t)$, $\tilde{\Phi} = \tilde{\Phi}(T;t)$.

Still, this periodicity alone does not lead to any change in climates: the system never jumps over T_2 (see fig).

If gaussian white noise, representing various mechanisms causing T fluctuations at a smaller time scale, is added to the system, yielding

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \tilde{F}(T;t) + \sigma\eta(t) \tag{1.8}$$

then for certain values of σ a jump between stable states becomes possible, with a frequency $\approx 2\pi/10^5$. That is, the system shows temperature oscillations at the forcing frequency, as noise allows the system to jump from one equilibrium to another (Figure 1.3). Note that oscillations arise from the joint effect of the periodic forcing and noise: noise pushes the system away from the equilibrium at every timestep, but only when the forcing asymmetrically tilts the double well potential such that the distance between the current equilibrium and the unstable one becomes small enough to allow the stochastic component to push the system over it, a climate transition (that is, a jump from T_1 to T_3 or viceversa) can occur. This can be seen as an "escape" from a stable equilibrium, requiring a jump over a (pseudo)energy potential "wall". This is a classic problem in physics, Kramer's rate [7] gives the frequency of jumps per second

$$r_K = \frac{\sqrt{V''(x_m)/m}\sqrt{|V''(x_b)/m|}}{2\pi\gamma} e^{-\frac{\Delta V}{D}} \tag{1.9}$$

where $V''(x_m)$ is the curvature around the minima, $V''(x_b)$ that at the maximum, ΔV the distance from the minimum to the maximum, and $D = \sigma^2/2$ the temperature of the system (noise intensity in our context). The inverse $\frac{1}{r_K}$ gives the residence time, the time spent at one of the minima. If our condition for SR is that it allows the system to follow the subthreshold forcing, then the optimal D is such that, given the forcing period T_ω and the residence time as a function of D, $T_K(D) = \frac{1}{r_K(D)}$ is

$$T_K(D) = \frac{T_\omega}{2} \tag{1.10}$$

Some useful considerations can be made on SR defined on the climate model. First, it requires a system containing a cubic nonlinearity, as in 1.2, such that 2 stable states separated by a potential wall exist. Additionally, the periodic forcing must be insufficient alone to cause a crossing (subthreshold). Later it will be shown that both of these assumptions can be relaxed. Last, as 1.10 suggests, too low D will not be able to push the system over the barrier at the frequency needed, and high enough D make the potential barrier effectively nonexistent, increasing r_K and making the double well model meaningless.

Hence, given a metric of how well the system follows the forcing, there will be a peak for a certain D. In this sense, there is a resonance in the system. For classic SR models, the natural measure is the spectral power (obtained by Fourier transform of the system response) at the forcing frequency relative to the neighboring ones: Signal-to-Noise Ratio (SNR) [8].

1.1.2 Small detour on dynamical systems

It will be shown that neurons are better modeled by 2D systems, due to the coexistence and interaction of 2 main ionic currents that govern a neuron's state $\left(\mathrm{Na}^+ \text{ and } \mathrm{K}^+\right)$, that determine a depolarization-hyperpolarization feedback loop. To analyze a 2+D systems the tools of dynamical systems theory are indispensable. A brief covering of the main methods will follow, as they will be referenced in the following chapters (for a more complete tractation, see [9]).

For any 2D system defined by

$$\dot{v} = f(v, u) \tag{1.11}$$

$$\dot{u} = g(v, u) \tag{1.12}$$

the system exists in the space $S = \{v\mathbf{e}_1 + u\mathbf{e}_2 \mid v, u \in \mathbb{R}\}$, defined by a set of parameters $\mathcal{P}$, and always identified with the state vector $\mathbf{s} = (v, u)$.

Then we define the mapping $F : \mathbb{R}^2 \to \mathbb{R}^2$ such that $F : \mathbf{s} \mapsto (f(\mathbf{s}), g(\mathbf{s}))$, the standard definition of a vector field (in $\mathbb{R}^n$). The direction of $F(\mathbf{s})$ is given by the sign of $f(\mathbf{s})$ and $g(\mathbf{s})$. So the curves defined by $f(v, u) = 0$ and $g(v, u) = 0$, the v- and u-nullclines, that separate the plane in changes in the sign of $\dot{v}$ and $\dot{u}$, are of particular interest. Note that $\dot{v} = \mathrm{d}v(t)/\mathrm{d}t$ and $\dot{u} = \mathrm{d}u(t)/\mathrm{d}t$. Then the evolution of the system's state from an initial position $\mathbf{s}(t_0) = (v(t_0), u(t_0))$ up to a time t, $\mathbf{s}(t)$, is governed by the flow of the vector field, formally $\varphi^t : S \to S$. Indeed, the trajectory $\varphi^{t-t_0}(\mathbf{s}_0)$ is a curve always tangent to F.

All points of intersection of the nullclines are such that $\dot{v} = \dot{u} = 0$. Consider the state $\mathbf{s}^*$ such that $f(\mathbf{s}^*) = g(\mathbf{s}^*) = 0$, if the system remains at $\mathbf{s}^*$ for all $t \geq 0$ then $\mathbf{s}^*$ is an equilibrium. Equilibria can be stable or unstable, the stability of an equilibria can be probed by analyzing the response of the system near $\mathbf{s}^*$ to a small disturbance $\tilde{\mathbf{s}} = \mathbf{s} - \mathbf{s}^*$,

such that $|\mathbf{s}^* - \mathbf{s}| \ll 1$. Note that by definition $F(\mathbf{s}^*) = \mathbf{0}$, so $\mathbf{s}^*$ is treated as a constant, and $\dot{\tilde{\mathbf{s}}} = F(\mathbf{s}^* + \tilde{\mathbf{s}})$. Then, in time domain, we are interested in $\varphi^t(\tilde{\mathbf{s}})$, for some time $t > t_0 = 0$.

However, the trajectory based on f, g usually cannot be calculated when they are nonlinear (as is the case for 1.18). Then we use the linearization of the system near the equilibria, solving

$$\dot{\tilde{\mathbf{s}}} = \mathrm{J}\tilde{\mathbf{s}} \equiv \begin{pmatrix} \dot{\tilde{v}} \\ \dot{\tilde{u}} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\partial f(\mathbf{s}^*)}{\partial v} & \frac{\partial f(\mathbf{s}^*)}{\partial u} \\ \frac{\partial g(\mathbf{s}^*)}{\partial v} & \frac{\partial g(\mathbf{s}^*)}{\partial u} \end{pmatrix}}_{\mathrm{J}} \begin{pmatrix} \tilde{v} \\ \tilde{u} \end{pmatrix} \tag{1.13}$$

note that in the linearization of the system quadratic and higher orders terms are omitted since they are evaluated on the small disturbance $\tilde{\mathbf{s}}$, and so are assumed to contribute extremely little to the solution.

To solve the system above we can use the eigenvalue method (if the eigenvectors are linearly independent) where solutions are in the form $\tilde{\mathbf{s}}(t) = e^{\lambda t}\boldsymbol{\xi}$, where $\boldsymbol{\xi}$ is the eigenvector. substituting into 1.13 we get

$$\dot{\tilde{\mathbf{s}}} = \mathrm{J}\tilde{\mathbf{s}} \to \lambda e^{\lambda t}\boldsymbol{\xi} = \mathrm{J}e^{\lambda t}\boldsymbol{\xi} \to \mathrm{J}\boldsymbol{\xi} = \lambda\boldsymbol{\xi} \tag{1.14}$$

The eigenvalues of J are found solving the characteristic equation $\det(\mathrm{J} - \lambda\mathrm{I}) = 0$, for 2×2 matrices, this leads to a quadratic polynomial that is solved as

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} \tag{1.15}$$

where $\tau = \mathrm{Tr}(\mathrm{J})$, $\Delta = \det(\mathrm{J})$, note that each eigenvalue will have its corresponding eigenvector, so our example system has $\boldsymbol{\xi}_1$, $\boldsymbol{\xi}_2$.

Finally, the eigenvectors of J form the eigenbase of the system. That is, their linear combination gives the linear approximation of $\varphi^t(\tilde{\mathbf{s}})$:

$$\dot{\tilde{\mathbf{s}}} = c_1 e^{\lambda_1 t - t_0}\boldsymbol{\xi}_1 + c_2 e^{\lambda_2 t - t_0}\boldsymbol{\xi}_2 \tag{1.16}$$

where c_1, c_2 are determined by $\tilde{\mathbf{s}}$.

Clearly, the behavior near the equilibria will be a linear combination of exponential motion along $\boldsymbol{\xi}_1$, $\boldsymbol{\xi}_2$ with direction determined by the signs of λ_1, λ_2. Each equilibria of a system can then be categorized by its eigenvalues.

The main types of equilibria relevant to this work are

- Nodes: real eigenvalues with the same sign. If $\lambda_{1,2} < 0$ the node is stable: trajectories converge to it. If $\lambda_{1,2} > 0$ unstable: trajectories diverge from it.
- Saddles: real eigenvalues with opposite sign. Thus always unstable.
- Focuses: complex-conjugate eigenvalues (when $\tau^2 - 4\Delta < 0$), trajectories spiral either inward or outward depending on stability. Sign of $\Re(\lambda_{1,2}) \neq 0$ determines the stability, while $\Im(\lambda_{1,2})$ determines the frequency of the rotation.

Knowing type and stability of equilibria, joined with knowing F, provides information on all possible behaviors of a system, given its initial state.

1.1.3 From Earth to neurons

Historically the first model used to study SR was that of 1.2, more generally expressed as a stochastic differential equation [10]

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x\left(a - bx^2\right) + \mu\gamma\cos(\omega t) + \sigma\eta(t) \tag{1.17}$$

where η is gaussian white noise, and $x(a - bx^2)$ cubic term, that for $a > 0$ has 3 fixed points. When defining the pseudo-potential as in 1.4 it gives the quartic polynomial that is the simplest realization of the double potential well (here ignoring forcing and noise component)

$$\begin{aligned}\Phi &= -\int -bx^3 + ax \\ &= \frac{b}{4}x^4 - \frac{a}{2}x^2 + C\end{aligned}$$

Intuitively, a neuron is a system that displays both quiescent and spiking behavior: that is, 2 stable states, separated by a "barrier" commonly called the voltage threshold. Note that the term threshold is hardly accurate: different current values can all elicit spikes in neurons depending on stimulation duration, some neurons respond to certain stimulation frequencies at currents lower than those needed to elicit a spike with a pulse-like stimulation, and others fire in response to hyperpolarizing currents (known as a rebound spike) [11, Chapter 7].

Unsurprisingly, a neuron can be modeled in the same form of 1.17 to show SR [12]. Interestingly, the same holds in more accurate neuron models.

1.1.4 The FitzHugh-Nagumo model

The FitzHugh-Nagumo (FHN) model [13, 14] is a 2D reduction of the Hodgkin-Huxley (HH) biophysical model [15], its attractiveness stems from being able to recreate the spiking behavior of neurons without the 4 differential equations that regulate the HH neuron, allowing for easier analysis of the model using the theory of dynamical systems.

The FHN model is defined by a fast and a slow variable

$$\frac{\mathrm{d}v}{\mathrm{d}t} = v\,(a - v)\,(v - 1) - w + I \tag{1.18}$$

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \varepsilon(bv - cw) \tag{1.19}$$

with v representing the (fast) membrane voltage and w the (slow, with $\varepsilon \ll 1$) feedback recovery mechanisms of the membrane (e.g., opening of K^+ channels). The main difference of this model with 1.2 is the second equation for w that allows for an additional behavior that cannot be seen in the climate model. However, a cubic term for $\dot{v}$ is present and one may wonder whether the resulting double well potential is a requirement for SR. It will be shown that it is not, and that it would describe an unrealistic regime for neural function under SR.

To see how the 2 models are related, both can be analyzed with the methods of 1.1.2. In the case of $\varepsilon \to 0$, we can treat w as a constant, then it is clear that the FHN model's phase portrait is topologically equivalent to 1.17 (see fig) for the right $\mathcal{P}$ (see B).

However, the FHN 1d reduction does not have a spiking mechanism, and the resulting neuron model would only show 2 resting states at different membrane voltages values v_1, v_3, exactly as in 1.2. Thinking about biological neurons, we would be describing a neuron with a single ionic current with a depolarizing effect, so voltage would only grow, with no recovery feedback. Clearly the 1d model is insufficient. Inspecting the full FHN we note that it is the slow w, entering as a subtractive term in v, that causes additional repolarization of the membrane, mapping to our biological understanding of the spiking mechanism.

Then the nonlinearity we are interested in is whichever is born from the interaction of the 2 opposing forces.

1.1.5 Neuron types and inputs

Different $\mathcal{P}$ construct different fields F, resulting in different equilibria, both number and type, and bifurcations; all leading to different behaviors. Thus the FHN can model different types of neurons, the main types are known as "integrators" (type-1) and "resonators" (type-2). The 2 types have different computational properties, but both can rightly be used for primary and secondary sensory neurons (which are of interest in this chapter), depending on the type of stimulus they are supposed to receive and process.

Stimuli can be modeled as instantaneous pulse-like or smooth ramp-up/down inputs. It is useful to clarify what is intended for instantaneous input (or shift in v), as it is also the formalization of a spike event in more simplified models, that will be used later. The common mathematical object used is Dirac's δ

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \tag{1.20}$$

such that

$$\int_{-\infty}^{\infty} \delta(x)dx = 1 \tag{1.21}$$

Then, defining the stimulation time as t_0, 1.18 becomes

$$\frac{\mathrm{d}v(t)}{\mathrm{d}t} = v\,(a - v)\,(v - 1) - w + I(t) \tag{1.22}$$

$$\frac{\mathrm{d}w(t)}{\mathrm{d}t} = \varepsilon(bv - cw) \tag{1.23}$$

where $I(t) = Q\delta(t - t_0)$, and Q is the voltage jump (more correctly Q/C, but the membrane's capacitance is assumed $C = 1$). Essentially $I(t)$ at t_0 shifts the whole system state vector to a new position in the state space $\tilde{\mathbf{s}} = (v + Q,\, w + 0)$, acting as a macro-perturbation from equilibrium.

It is to note that the δ-like input is one limited way to model the stimulation that a neuron may be subjected to. Often a neuron may receive multiple inputs from pre-synaptic elements that can vary in intensity, depending on many biological details. Thus, another way to model stimulation is to allow $I(t)$ to vary continuously with time, mimicking the smoothing summation of incoming inputs at different timings.

From the perspective of dynamical systems, the effect is not of a perturbation on $\mathbf{s}$. Rather, when the changes in I are slow enough, the geometry of the phase portrait itself changes, as w changes accordingly. Effectively I is to be thought of as another parameter in $\mathcal{P}$ that we choose to vary.

As seen in 1.1.2, this is the setting for a codimension-1 bifurcation. Then, unlike δ inputs, smooth ones result in a change in the topology of the phase portrait of the system, changing its overall behavior.

In 1.1.1 we saw that SR happens for an input $\mu(t)$ that was unable by itself to lift the system over the pseudopotential "wall" (intuitively no better tracking of $\mu(t)$ could be made otherwise). It is clear that for neurons the "wall" is the difference between the spiking and non-spiking state. Therefore, it is now necessary to clarify the meaning of threshold.

1.1.6 Neural nonlinearity, or threshold phenomena

The usual working definition of a threshold is a membrane voltage value v_ϑ that when reached is followed by a large excursion in v, while no excursion is present for $v < v_\vartheta$. The discontinuity in the trajectories of v near v_ϑ is our wall to be crossed. However, looking at

the vector fields depicted in the phase portraits, it is unclear whether there really is such a threshold value.

Indeed, consider the FHN in its simplest configuration: having a unique globally attracting stable equilibrium, this means that any excitation of the system will eventually converge to it. This mirrors our understanding of a neuron with a preferred resting state that after stimulation relaxes back to it.

The FHN system so defined does not have any discontinuity in its vector flow, no all-or-none response can arise theoretically (see C for an argument). In fig [fig monstable] such a system is depicted. The phase portrait is divided by the w/v-nullclines in 4 regions, depending on the direction of $(\dot{v}, \dot{w})$, a δ-like stimulus may push the system state vector $\mathbf{s}$ in either the $(\dot{v} < 0, \dot{w} > 0)$ or the $(\dot{v} > 0, \dot{w} > 0)$ regions. In the former case $\mathbf{s}$ will quickly return to $\mathbf{s}_0$, while in the latter it will go quickly to the right of the unstable middle branch of the v-nullclines and cross it after its right knee, then coming back to $\mathbf{s}_0$ as $\dot{v} < 0$.

However, for the right stimulation, the system may also travel along the middle branch of v-nullcline up to the right knee before jumping to a stable branch. infact, there exist a set of those intermediate trajectories (canards), also called the slow manifold, as at the v-nullcline $\dot{v} = 0$, and the dynamics are governed by the slow w variable.

Variable amplitude action potential are also present in biological neurons [16], but the overwhelming majority of observations points to single threshold values and all-or-none responses. This is due to the thinness of the slow manifold: given the unstability of the middle branch few trajectories follow it for enough time to produce the intermediate response. Indeed, The size of the interval is of order $O(e^{-c/\varepsilon})$, where c is a positive constant and $\varepsilon \ll 1$ is the separation between the fast and slow time scales [17]. Then in the physical world channel noise alone would make it impossible to remain on the slow manifold, that can be regarded in realistic conditions as a "quasi-threshold" or a threshold manifold [18].

Monostable models are usually near an Hopf bifurcation, hence resonators. Integrators are instead modeled by systems near saddle-node bifurcations, then at least a node and a saddle must exist in the state space. Assuming a δ input that shifts the state vector $\mathbf{s}$ from the stable node (again, the resting state), if Q moves $\mathbf{s}$ to the left of the stable branch of the saddle point, $\mathbf{s}$ trajectory is first dominated by the negative eigenvalue, then by then positive, being repelled back to the node. That is, no large excursion happens. When Q is large enough to move $\mathbf{s}$ to the right of the stable branch, the system follows again the eigenvector with the negative eigenvalue, but is then repelled to the right, resulting in a large excursion in v (see Fig [saddle-node fig]). Therefore in integrators there exist a real discontinuity: the stable branch of the saddle, called the separatrix, represents a set of thresholds [19].

For non δ stimulation the threshold may be viewed as the critical value of the codim-1 parameter (i.e., the input voltage), whenever it is crossed the system undergoes a bifurcation that changes its behavior. Usually in FHN and similar models this leads to a continued spiking state, with frequency dependent on the input for saddle-node and a roughly stable instrinsic frequency in hopf bifurcations.

Various models and (quasi)threshold phenomena have been described, returning to the arguments in 1.1.1, it is clear that nonlinear elements in neurons create different types of nonlinearity in responses. However, once biological parameter ranges are taken into consideration, the difference between a spike and a non-spike response is what constitues our "wall" to be crossed. Note that the different dynamical properties of the different models have an important role in the neurocomputational characteristics of neurons, but for the purpose of studying SR for recovery of an external input that simple concept is all we need. Indeed, SR is found in different dynamical configurations of the FHN [20–22].

This guides us towards a useful generalization, that will be made evident when con-

sidering the setting of application of SR in neural systems.

1.1.7 Role of SR in sensory processing

In the climate model noise allowed the periodic changes of total incoming solar radiation that were too weak to move the earth systems temperatures alone, to influence it. This problem naturally maps to signal detection in neural systems: a signal too weak to be detected could be with the right noise level.

Appendix A

Stability of Equilibria

In Benzi et al. [5]

Appendix B

Phase portraits of climate and FHN models

Appendix C

No threshold in monostable FHN

Consider the FHN system in a configuration having a unique globally attracting stable equilibrium $\mathbf{s}_0 = (v_0, w_0)$.

Take $\dot{\mathbf{s}} = f(\mathbf{s})$ as

$$f(v,w) = \left(v - \frac{v^3}{3} - w - I, \varepsilon(v + a - bw)\right) \tag{C.1}$$

then $f : \mathbb{R}^2 \to \mathbb{R}^2$ is smooth on R^2. We define a stimulation function continuous on a compact set of inputs values $Q = [0, q_{max}] \subset \mathbb{R}$, modeling an instantaneous stimulation

$$\gamma : Q \to \mathbb{R}^2,\ q \mapsto (v_0 + q, w_0) \tag{C.2}$$

then $A = \gamma(Q) \subset \mathbb{R}^2$ is the compact set of starting states after stimulation. We are interested in the properties of the flow originating from A for all $t \geq 0$.

By definition of a stable equilibrium, for any neighborhood $U \supset \mathbf{x}_0$ there exist a $V \supset \mathbf{x}_0$ such that $\varphi^t(\mathbf{x}) \in U$ for all $\mathbf{x} \in V$ and $t > 0$. That is, once entered in V, the orbit never leaves U (Lyupanov stability). Because of global attraction of $\mathbf{s}_0$, it is guaranteed that there exist a time for which $\varphi^t(\mathbf{a})$ is inside any $W \supset \mathbf{s}_0$. Then we take $\mathrm{T}^e(B(\mathbf{a}))$ as the time in which all $\varphi^t(B(\mathbf{a})) \in V$ for a sufficiently small neighborhood $B(\mathbf{a}) \supset \mathbf{a}$. Since A is compact, there are always finitely many such $B(\mathbf{a}) \in A$.

Hence we can define

$$\mathrm{T}^* = \max\left\{\mathrm{T}^e(B(\mathbf{a}_1)), \ldots, \mathrm{T}^e(B(\mathbf{a}_n))\right\} \tag{C.3}$$

as the time at which all orbits starting at A are guaranteed to be in U, it follows that $\{\varphi^t(A)\}_{t\in[0,\mathrm{T}^*]}$ is a bounded set of continuous trajectories. For Lyupanov stability, once entered V all orbit stay in U for all time, and for asymptotic stability of $\mathbf{s}_0$ all $\varphi^t(\mathbf{x}) \to \mathbf{s}_0$ as $t \to \infty$. So $\{\varphi^t(A)\}_{t>\mathrm{T}^*}$ is bounded.

Now we consider the complete flow from A as the closed set

$$C = \{\varphi^t(A)\}_{t\in[0,\mathrm{T}^*]} \cup \{\varphi^t(A)\}_{t>\mathrm{T}^*} \tag{C.4}$$

since both elements are bounded, C is a closed and bounded set in $\mathbb{R}^2$, so compact.

Therefore smooth f is Lipstichz continuous on C, that is, there exist a uniform L such that

$$|f(x) - f(y)| = L\,|x - y| \tag{C.5}$$

for all $x, y \in C$, and Gronwall's inequality applies

$$|\varphi^t(x) - \varphi^t(y)| = e^{Lt}\,|x - y| \tag{C.6}$$

for all $x, y \in C$. Thus for any closed interval in A, the resulting flow is always continuous.

To see the consequences of this: let $v(\mathbf{s}, t)$ be the function giving the v value of the system state vector at any t. Then on any arbitrary closed interval $\mathcal{I} = [0, T]$ we take $\mathbf{s}_t = \varphi^t(\mathbf{a})$ for $\mathbf{a} \in A$. We can define the maximum v for a given trajectory in $\mathcal{I}$

$$V_m(\mathbf{a}) = \max_{t \in \mathcal{I}} \{v(\varphi^t(\mathbf{a}), t)\} \tag{C.7}$$

Considering all possible trajectories in C, C.6 applies:

$$|V_m(\mathbf{a}_1) - V_m(\mathbf{a}_2)| = e^{Lt} |\mathbf{a}_1 - \mathbf{a}_2| \tag{C.8}$$

for all $\mathbf{a}_1$, $\mathbf{a}_2 \in A$, in the arbitrary interval $\mathcal{I}$.

That is, the maximum voltage of the spike upstroke is always continuous for all stimulation values: no discontinuity in the voltage trace over time exists.

Note that as $t \to \infty$ the exponential could indeed lead to a great distance from v values. However, the maximum of a spike always happens in a finite interval after the stimulation, or it does not.

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